

Exploring further key uncertainties about the sardine assessment

C.L. de Moor*

Correspondence email: carryn.demoor@uct.ac.za

Some of the key uncertainties about the current assessment of South African sardine are explored including assuming time varying and density dependent natural mortality, fitting to parasite prevalence-at-length data, and assuming the catch by stock off the west coast is proportional to the exploitable numbers-at-age by stock. These results should assist in the selection of a baseline operating model(s) for the joint sardine-anchovy Management Strategy Evaluation.

Keywords: assessment, density dependence, natural mortality, parasite prevalence, robustness tests, sardine, South Africa

Introduction

This document explores some of the key uncertainties about the current assessment of South African sardine (de Moor 2025a,b,d) to assist in the selection of a baseline operating model(s) for the joint sardine-anchovy Management Strategy Evaluation.

Methods

These analyses are based on those of de Moor (2025a,b), but with fixed values for $M_{s,y,0}^{WTS}$, $M_{s,y,1+}^{WTS}$ and M_{inc}^{WTS} (de Moor 2025d). The baseline model assumes $M_{w,y,0}^{CTS} = 1.0$, $M_{w,y,1+}^{CTS} = 0.8$, $M_{s,y,0}^{WTS} = 1.25$, $M_{s,y,1+}^{WTS} = 1.25$ and assumes time-invariant growth, i.e. $\eta_{c,y}^k = 0$. A beta prior is used for $2(t_{0,c} + 0.25) \sim \text{Beta}(1.05, 1.05)$, thereby restricting $-0.25 \leq t_{0,c} \leq 0.25$. The data are weighted as per the likelihood equations in Appendix A of de Moor (2025a), but without fitting to the parasite prevalence-at-length data. Catch by stock off the west coast is in proportion to the exploitable numbers-at-age by stock, i.e. $\Phi_1 = \Phi_2 = 1$. Some further modifications to prior distributions of other parameters or fixing of ‘nuisance’ parameters for the hybrid approach to MCMC may be required in due course if data are found to be uninformative w.r.t. reliably estimating those parameters.

Further key uncertainties require testing before the baseline operating model(s) for the joint sardine-anchovy Management Strategy Evaluation can be finalised. Results presented in this document do not estimate a stock-recruitment relationship during conditioning (de Moor 2025a,b).

Given natural mortality now varies between stocks and can vary between coasts too, a range of sensitivity tests were run to consider whether annually varying natural mortality has a substantial effect, and if so, for which stocks/areas (Table 1).

Density dependent natural mortality was tested, with $\bar{M}_{c,y,a}^j = \bar{M}_{c,a}^j + e^{-\chi B_{c,y-1}^S}$, $\chi > 0$ (models S_{DD1} , S_{DD2} , S_{DD3}). The relationship is assumed to be with total biomass by coast as might be influenced by predation on sardine. One model (S_{DD4}) was also run assuming coast-dependent relationships.

The impact of additionally fitting to parasite prevalence at length data while estimating $I_y \sim \text{Beta}(1.05, 1.05)$ was tested with south coast data east of 22° only (S_{prev}) and east of 20° (S_{prev2}).

* MARAM (Marine Resource Assessment and Management Group), Department of Mathematics and Applied Mathematics, University of Cape Town, Rondebosch, 7701, South Africa.

Finally, five options for allowing the catch by stock off the north-west and south-west coasts to differ from the proportions of exploitable numbers-at-age by stock are considered:

S_{SW-WTS} - Catch off the south west coast is weighted towards WTS, with $\Phi_1 = 10$.

S_{SW-CTS} - Catch off the south west coast is weighted towards CTS, with $\Phi_1 = 0.1$.

S_{NW-CTS} - Catch off the north west coast is weighted towards CTS, with $\Phi_2 = 10$.

S_{NW-WTS} - Catch off the north west coast is weighted towards WTS, with $\Phi_2 = 0.1$.

S_{SW-NW} - Catch off the north west coast is weighted towards CTS, and catch off the south west coast is weighted towards WTS, with $\Phi_1 = \Phi_2 = 10$.

Results and Discussion

Only a small improvement in the fit to the data is achieved by allowing juvenile natural mortality to vary annually (S_{Mj}), while there is a substantial improvement in the fit to the data when adult natural mortality is allowed to vary (S_{Mad}) (Table 2). This is consistent with the findings of de Moor (2020). Also consistent with the findings of de Moor (2020) is that a higher natural mortality rate is estimated since approximately 2004/6 compared to the first half of the time series (Figures 1-2) and the autocorrelated changes in natural mortality may suggest that this model is not simply estimating changes to natural mortality to ‘fit to noise’ (Smith *et al.* 2011, de Moor 2020). Figure 2c indicates that inter-annual changes in natural mortality in WTS, and particularly in WTS off the south coast – is more likely / preferred to inter-annual changes in CTS natural mortality.

When natural mortality was assumed to be density dependent, two local minima were found (i) an increase in M in more than 1 decimal place was only estimated below 10 000t of model estimated biomass, i.e. $\bar{M}_{c,y,a}^j = \bar{M}_{c,a}^j + e^{-0.3B_{c,y}^S} (S_{DD1})$ and (ii) an increase in M below 1 000 000t of model estimated biomass, i.e. $\bar{M}_{c,y,a}^j = \bar{M}_{c,a}^j + e^{-0.0047B_{c,y}^S} (S_{DD3})$. The latter produced a better fit to the data than the baseline model with time invariant natural mortalities (Table 3). A third alternative with $\chi = 0.01$ was also run. When density dependence was assumed to be coast-dependent, the west coast natural mortality increased below $\sim 400\,000$ t model estimated biomass ($\chi_w = 0.009$), while the increase off the south coast was more gradual ($\chi_s = 0.001$). The density dependent relationships are shown in Figures 3 and 4. Consistent with the results from above when inter-annual variability in M was assumed, the natural mortality rate off the west coast is estimated to be higher during the second half of the time series, with a peak in the late 2000s (Figure 5). However, the natural mortality rate off the south coast is different, with lower values estimated in the early 2000s (Figure 5). The alternative of natural mortality increasing with biomass (e.g. competition for food, $\bar{M}_y = \bar{M} + e^{\chi B_y - 1}$) was not tested, given the higher natural mortality rates estimated in recent years when biomass has been low (Figures 1 and 2).

The contribution to the likelihood between S_{prev} and S_{prev2} cannot be directly compared due to the different south coast data used for the two tests. Figures 6 to 12 show the difference in model fit between S_0 , S_{prev} and S_{prev2} .

The results in this document when assuming the catch in the north-west and south-west regions is not necessarily split between stocks in proportion to the exploitable numbers available by stock (Table 3) are consistent with those obtained by de Moor (2025e), with an improved fit to the data obtained only for S_{SW-CTS} .

Given these results, the following models are suggested to be retained, though discussion about which will form part of a reference set of baseline models or only as a robustness test is required:

- S_{Mad1} - Annually varying adult natural mortality in WTS off both coasts, i.e. random effects model with $\sigma_{ad} = 0.2$ and $\rho \sim U(0,1)$.
- S_{Mad2} - Annually varying adult natural mortality in WTS off the south coast, i.e. random effects model with $\sigma_{ad} = 0.2$ and $\rho \sim U(0,1)$.
- S_{DD1} - Coast-independent density dependent natural mortality, with $\bar{M}_{c,y,a}^j = \bar{M}_{c,a}^j + e^{-\chi B_{c,y}^S}, \chi > 0$.
- S_{DD2} - Coast-dependent density dependent natural mortality, with $\bar{M}_{c,y,a}^j = \bar{M}_{c,a}^j + e^{-\chi_c B_{c,y}^S}, \chi > 0$.
- S_{prev} - Include parasite prevalence-at-length data in the likelihood using samples off the south coast east of 22°E and estimate $I_y \sim \text{Beta}(1.05, 1.05)$.
- S_{prev2} - Include parasite prevalence-at-length data in the likelihood using samples off the south coast east of 20°E and estimate $I_y \sim \text{Beta}(1.05, 1.05)$.
- S_{SW-WTS} - Catch off the south west coast is weighted towards WTS, with $\Phi_1 = 10$.
- S_{SW-CTS} - Catch off the south west coast is weighted towards CTS, with $\Phi_1 = 0.1$.

The latter two are suggested as two extremes, noting S_{SW-WTS} was preferred a priori by the industry, but S_{SW-CTS} gives a better fit to the data.

Work towards which assumed stock-recruitment relationship (if any) is to be estimated during or after conditioning for use in baseline models or robustness tests for the operating models is continuing in parallel. These are focusing on a Beverton Holt relationship and a ‘binning’ approach with no stock-recruitment relationship. It is suggested that the Ricker relationship not be considered and that a Hockey Stick relationship could be considered as a robustness test.

Further robustness tests which have previously been requested/suggested for key uncertainties about the sardine population dynamics are listed below:

- S_{M1} - $M_{w,y,0}^{CTS} = 1.4, M_{w,y,1+}^{CTS} = 0.8, M_{s,y,0}^{WTS} = 1.40, M_{s,y,1+}^{WTS} = 1.07$ (de Moor 2025d).
- S_{M1} - $M_{w,y,0}^{CTS} = 0.8, M_{w,y,1+}^{CTS} = 0.8, M_{s,y,0}^{WTS} = 1.05, M_{s,y,1+}^{WTS} = 1.05$ (de Moor 2025d).
- S_{Minc} - $M_{inc}^{WTS} = 0.2$ or $M_{inc}^{WTS} = 0.5$ (de Moor 2025d).
- S_{K1} - Time-variation in growth by cohort with $\eta_y^K \sim N(0, 0.2^2)^1$.
- S_{K2} - Time-variation in growth by year with $\eta_y^K \sim N(0, 0.2^2)$.
- S_{k0.6} - Acoustic survey bias fixed, $k_{ac}^S = 0.6$.
- S_{k1} - Acoustic survey bias fixed, $k_{ac}^S = 1.0$.
- S_{lamR} - Additional variance (over and above the survey sampling CV) associated with the recruit survey, fixed $\lambda_r^2 = 0.02$.
- S_{lamN} - Additional variance (over and above the survey sampling CV) associated with the November survey, fixed $\lambda_N^2 = 0.02$.
- S_{sur} - standard deviation associated with the survey proportion-at-length data is estimated annually (through closed form solution and bounded between [0.04,0.1]) and not as a time-invariant standard deviation.

¹ There was little improvement in the fit to the data for additionally allowing the growth to vary by coast (de Moor 2025e).

- S_{com} - standard deviation associated with the commercial proportion-at-length data is estimated annually (through closed form solution and bounded between [0.04,0.1]) and not as a time-invariant standard deviation.
- S_{down} - Down weight the survey and commercial proportion-at-length data, i.e. w_{propl}^{sur} and w_{propl}^{com} in equations A34 and A35 (de Moor 2025a) replaced with $0.3w_{propl}^{sur}$ and $0.3w_{propl}^{com}$.
- S_{mat} - Time-invariant maturity ogive.
- S_{weight} - Annual length-weight relationships are used for 2005-2022, with the average time-invariant relationship in other years (SWG-PEL, November 2023)
- S_{2020} - Upweight the November 2020 survey estimate of biomass off the south coast to ensure residuals are unbiased.

de Moor (2020) also considered two sensitivity tests to the timing of the movement and infection, but the timing of the movement and infection is now modelled to occur mid-way through the 3rd quarter. de Moor (2020) also included a sensitivity test not assuming a 5+ plus group, but results were insensitive to this changed assumption.

Possible robustness tests that would only include changes in the future dynamics of the resource include:

- S_{pulse} - Pulses of three consecutive years of above-average sardine recruitment are modelled to occur with 1/40 chance.
- S_{low} - Three consecutive years of below-average recruitment are assumed for 2030-32. (A change in future carrying capacity is typically considered in MSEs to model a change in productivity, but this is only possible if a stock recruitment relationship is assumed).
- S_{move} - Alternative assumptions for future active movement of WTS from the west to the south coast (to be determined after baseline model(s) finalised).
- S_p - Alternative assumptions for future proportions of WTS recruitment which recruit to the west coast (to be determined after baseline model(s) finalised).

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Table 1. Alternative options considered for allowing interannual variability in the natural mortality rate.

Name	$M_{w,y,0}^{CTS}$	$M_{w,y,1+}^{CTS}$	$M_{s,y,0}^{CTS}$	$M_{s,y,1+}^{CTS}$	$M_{w,y,0}^{WTS}$	$M_{w,y,1+}^{WTS}$	$M_{s,y,0}^{WTS}$	$M_{s,y,1+}^{WTS}$	σ_j	σ_{ad}
S ₀	1.0	0.8	1.0	0.8	1.25	1.25	1.25	1.25		
S ₁	Vary	0.8	Vary	0.8	Vary	1.25	Vary	1.25	0.1	-
S ₂	Vary	0.8	Vary	0.8	Vary	1.25	Vary	1.25	0.2	-
S ₃	Vary	0.8	Vary	0.8	1.25	1.25	1.25	1.25	0.1	-
S ₄	Vary	0.8	Vary	0.8	1.25	1.25	1.25	1.25	0.2	-
S ₅	1.0	0.8	1.0	0.8	Vary	1.25	Vary	1.25	0.1	-
S ₆	1.0	0.8	1.0	0.8	Vary	1.25	Vary	1.25	0.2	-
S ₇	1.0	0.8	1.0	0.8	1.25	1.25	Vary	1.25	0.2	-
S ₈	1.0	0.8	1.0	0.8	Vary	1.25	1.25	1.25	0.2	-
S _{1a}	1.0	Vary	1.0	Vary	1.25	Vary	1.25	Vary	-	0.1
S _{2a}	1.0	Vary	1.0	Vary	1.25	Vary	1.25	Vary	-	0.2
S _{3a}	1.0	Vary	1.0	Vary	1.25	1.25	1.25	1.25	-	0.1
S _{4a}	1.0	Vary	1.0	Vary	1.25	1.25	1.25	1.25	-	0.2
S _{5a}	1.0	0.8	1.0	0.8	1.25	Vary	1.25	Vary	-	0.1
S _{6a}	1.0	0.8	1.0	0.8	1.25	Vary	1.25	Vary	-	0.2
S _{7a}	1.0	0.8	1.0	0.8	1.25	1.25	1.25	Vary	-	0.2
S _{8a}	1.0	0.8	1.0	0.8	1.25	Vary	1.25	1.25	-	0.2

Table 2. The contributions to the objective function from likelihood (equations A29 – A33 of de Moor 2025a) and prior components for the model without a stock recruitment relationship together with some key parameter estimates at the joint posterior mode. Results are shown for different annually varying natural mortality rates listed in Table 1.

		S ₀	S ₁	S ₂	S ₃	S ₄	S ₅	S ₆	S ₇	S ₈	S _{1a}	S _{2a}	S _{3a}	S _{4a}	S _{5a}	S _{6a}	S _{7a}	S _{8a}
Objective function		-855.0	-912.1	-884.6	-911.8	-883.4	-912.0	-884.5	-883.7	-883.6	-915.0	-891.4	-912.5	-885.4	-914.4	-890.2	-888.9	-884.0
log likelihood	$-\ln L$	-904.0	-904.7	-906.7	-904.0	-904.1	-904.6	-906.6	-904.8	-904.6	-909.7	-916.9	-905.4	-907.0	-908.8	-914.8	-913.3	-905.2
	$-\ln L^{Nov}$	41.9	41.8	41.7	41.9	41.9	41.8	41.7	41.8	41.8	41.1	40.1	41.9	42.0	40.7	39.1	39.7	41.5
	$-\ln L^{rec}$	30.7	30.7	30.5	30.7	30.7	30.6	30.4	30.7	30.6	30.0	29.3	30.6	30.5	29.9	29.1	29.5	30.4
	$-\ln L^{sur\ prop}$	-495.6	-496.1	-497.6	-495.6	-495.7	-496.0	-497.5	-496.2	-495.8	-498.7	-502.8	-496.6	-498.1	-497.4	-499.9	-499.5	-496.1
	$-\ln L^{com\ prop}$	-481.1	-481.1	-481.2	-481.1	-481.1	-481.1	-481.2	-481.1	-481.1	-482.2	-483.5	-481.3	-481.5	-481.9	-483.1	-483.0	-481.1
log priors	$\ln(k_{ac}^S)$	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.6	-0.6	-0.5	-0.5	-0.5	-0.5
	$move_{y,1}$	4.1	4.1	4.1	4.1	4.1	4.1	4.1	4.1	4.1	4.0	4.0	4.1	4.1	4.0	4.0	4.0	4.1
	$\bar{l}_{1,y}$	41.9	41.9	41.8	41.9	41.9	41.9	41.8	41.9	41.9	41.8	41.8	41.8	41.8	41.8	41.8	41.9	41.9
	ϑ_a	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
	S_{50}	2.6	2.6	2.8	2.6	2.6	2.6	2.8	2.7	2.6	2.6	2.8	2.6	2.6	2.6	2.7	2.7	2.6
	$t_{0,c}$	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
	η_y^M	-	56.4	27.1	56.7	28.3	56.4	27.1	27.9	28.0	54.3	23.6	56.1	27.2	54.6	24.4	24.7	27.7
	k_{ac}^S	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.68	0.67	0.70	0.72	0.69	0.68	0.68	0.69
	$k_{w,r}$	0.29	0.29	0.30	0.29	0.29	0.29	0.29	0.29	0.30	0.30	0.31	0.30	0.32	0.30	0.30	0.29	0.30
	$k_{s,r}$	0.29	0.29	0.29	0.29	0.29	0.29	0.29	0.29	0.29	0.30	0.31	0.30	0.32	0.30	0.29	0.29	0.29
	ρ	-	0.85	0.88	0.99	0.99	0.85	0.80	0.99	0.84	0.86	0.78	0.99	0.99	0.84	0.81	0.84	0.82

Table 3. The contributions to the objective function from likelihood (equations A29 – A33 of de Moor 2025a) and prior components for the model without a stock recruitment relationship together with some key parameter estimates at the joint posterior mode. Results are shown for models detailed in the main text.

		S ₀	S _{DD1}	S _{DD2}	S _{DD3}	S _{DD4}	S _{prev}	S _{prev2}	S _{SW-WTS}	S _{SW-CTS}	S _{NW-CTS}	S _{NW-WTS}	S _{SW-NW}
Objective function		-855.0	-855.1	-854.9	-855.6	-861.3	2174.1	2531.8	-853.0	-855.4	-855.2	-852.6	-852.1
log likelihood	$-\ln L$	-904.0	-904.0	-904.0	-904.8	-911.4	-846.6	-862.9	-903.1	-904.0	-903.9	-902.1	-902.4
	$-\ln L^{Nov}$	41.9	41.9	40.8	40.4	38.9	48.6	46.8	41.8	42.0	42.0	42.0	42.1
	$-\ln L^{rec}$	30.7	30.7	30.4	29.6	28.8	45.4	43.6	30.5	30.5	30.4	28.9	29.1
	$-\ln L^{sur\ prop1}$	-495.6	-495.6	-493.4	-493.1	-497.1	-471.5	-479.3	-495.2	-494.4	-495.0	-491.5	-493.6
	$-\ln L^{com\ prop1}$	-481.1	-481.1	-481.8	-481.6	-482.0	-469.1	-473.9	-480.2	-482.1	-481.3	-481.5	-480.1
	$-\ln L^{prev}$	-	-	-	-	-	2966.6	3341.7	-	-	-	-	-
log priors	$\ln(k_{ac}^S)$	-0.5	-0.5	-0.6	-0.6	-0.6	-0.6	-0.1	-0.3	-0.5	-0.5	-0.3	-0.5
	$move_{y,1}$	4.1	4.1	4.1	3.9	4.1	5.5	5.5	4.0	3.9	4.0	3.9	4.3
	$\bar{l}_{1,y}$	41.9	41.9	41.8	41.9	42.2	42.6	41.8	42.7	41.7	41.5	42.2	42.4
	ϑ_a	0.8	0.8	0.7	0.8	0.8	0.9	0.8	0.8	0.8	0.8	0.9	0.8
	S_{50}	2.6	2.6	2.9	2.9	3.5	3.4	2.6	2.5	2.5	2.6	2.4	2.9
	$t_{0,c}$	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.3	0.2
	I_y	-	-	-	-	-	2.1	2.1	-	-	-	-	-
	k_{ac}^S	0.69	0.69	0.72	0.69	0.69	0.87	0.60	0.64	0.69	0.69	0.64	0.67
	$k_{w,r}$	0.29	0.29	0.29	0.26	0.25	0.38	0.25	0.27	0.29	0.29	0.28	0.28
	$k_{s,r}$	0.29	0.29	0.29	0.26	0.25	0.31	0.25	0.27	0.29	0.29	0.28	0.28

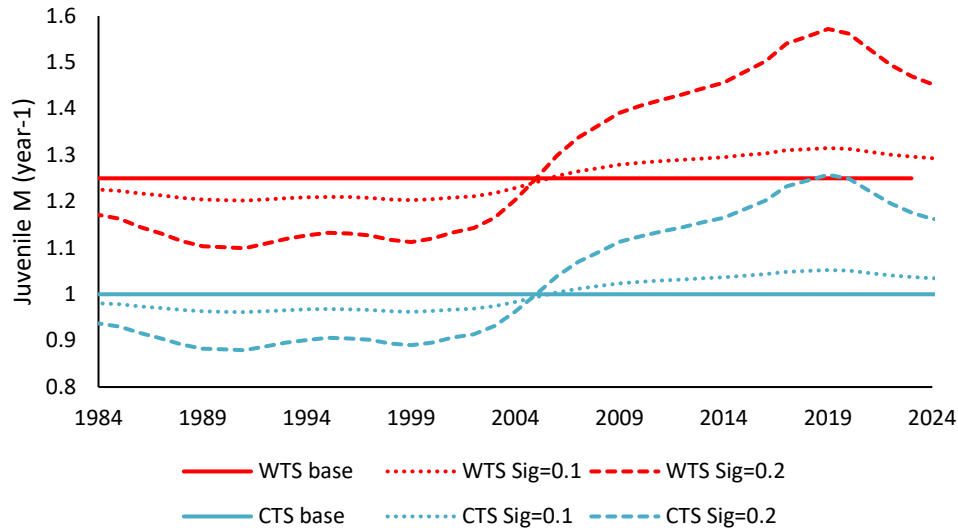


Figure 1a. The annual juvenile natural mortality rate for WTS (red) and CTS (blue) for S_0 , S_1 and S_2 .

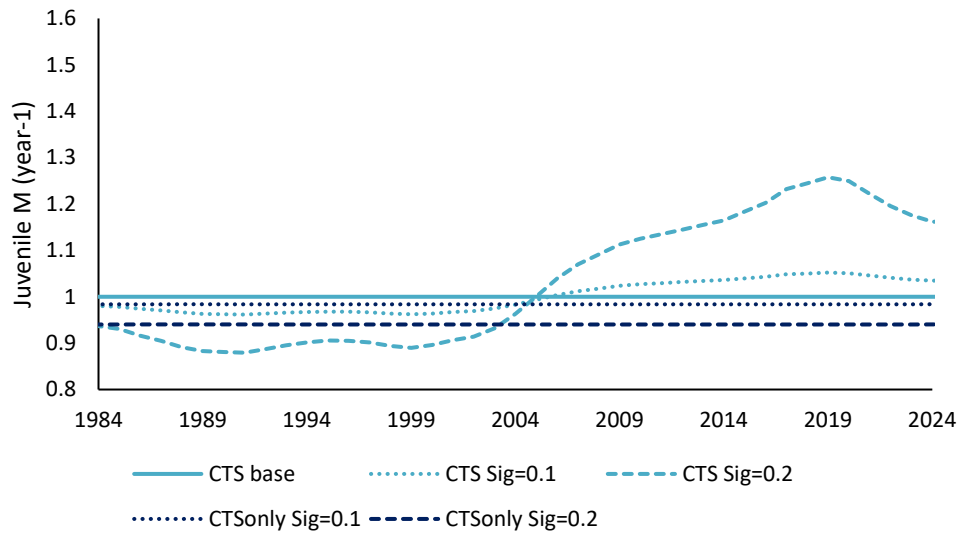


Figure 1b. The annual juvenile natural mortality rate for CTS for S_0 , S_1 , S_2 , S_3 and S_4 .

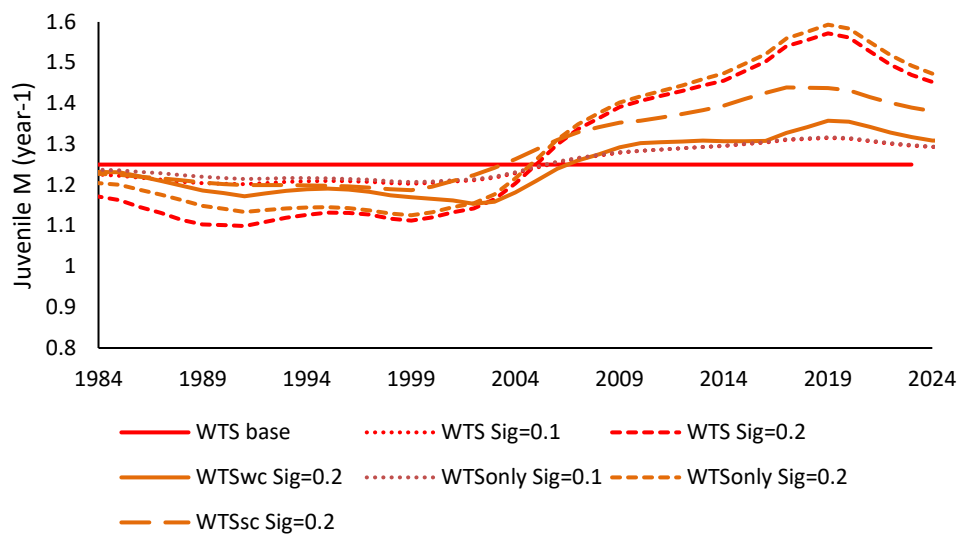


Figure 1c. The annual juvenile natural mortality rate for WTS for S_0 , S_1 , S_2 , S_5 , S_6 , S_7 and S_8 .

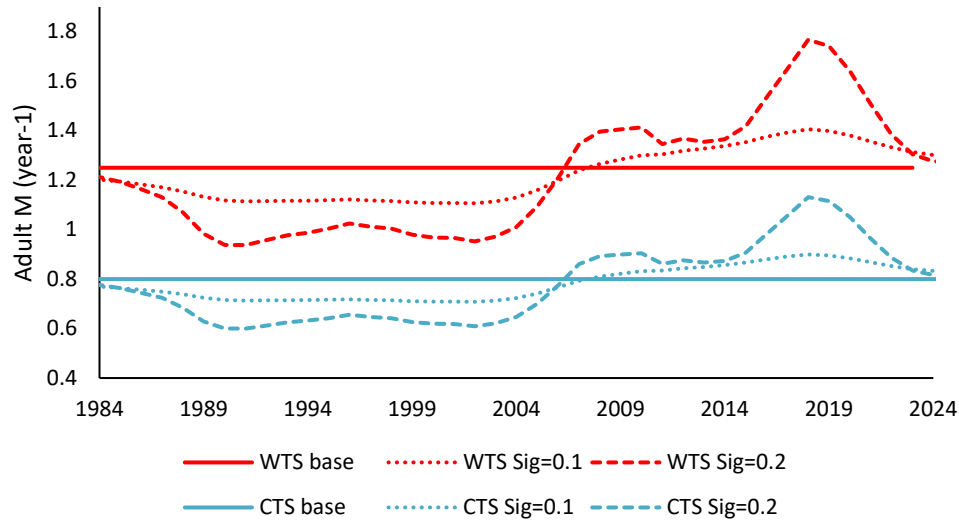


Figure 2a. The annual adult natural mortality rate for WTS (red) and CTS (blue) for S_0 , S_{1a} and S_{2a} .

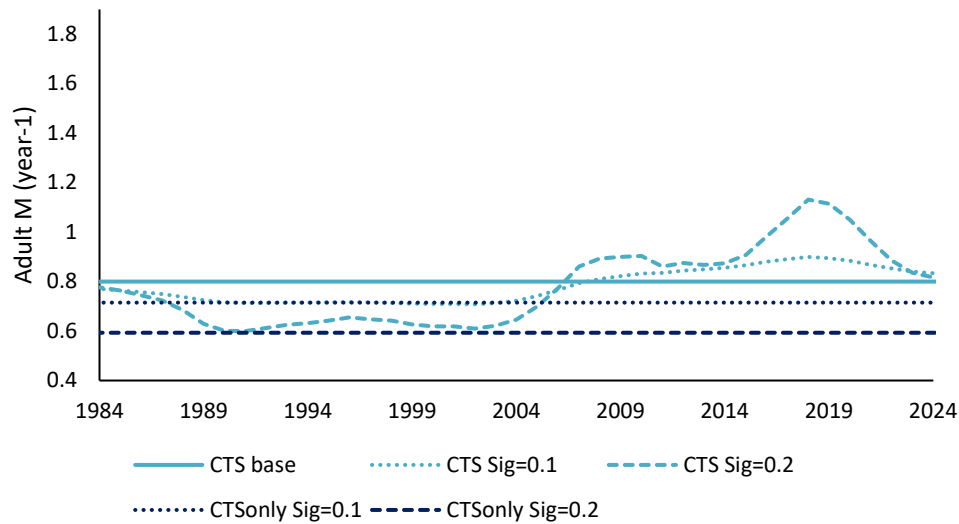


Figure 2b. The annual adult natural mortality rate for CTS for S_0 , S_{1a} , S_{2a} , S_{3a} and S_{4a} .

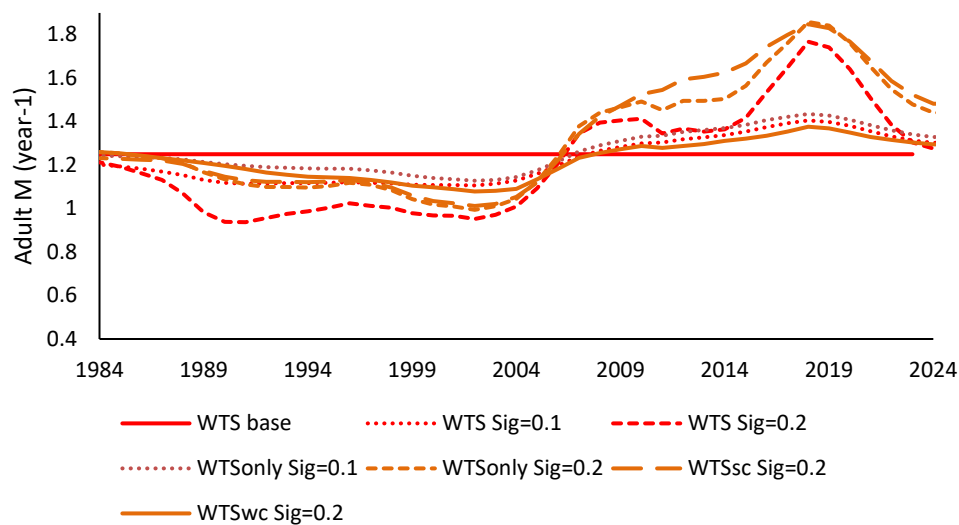


Figure 2c. The annual adult natural mortality rate for WTS for S_0 , S_{1a} , S_{2a} , S_{5a} , S_{6a} , S_{7a} and S_{8a} .

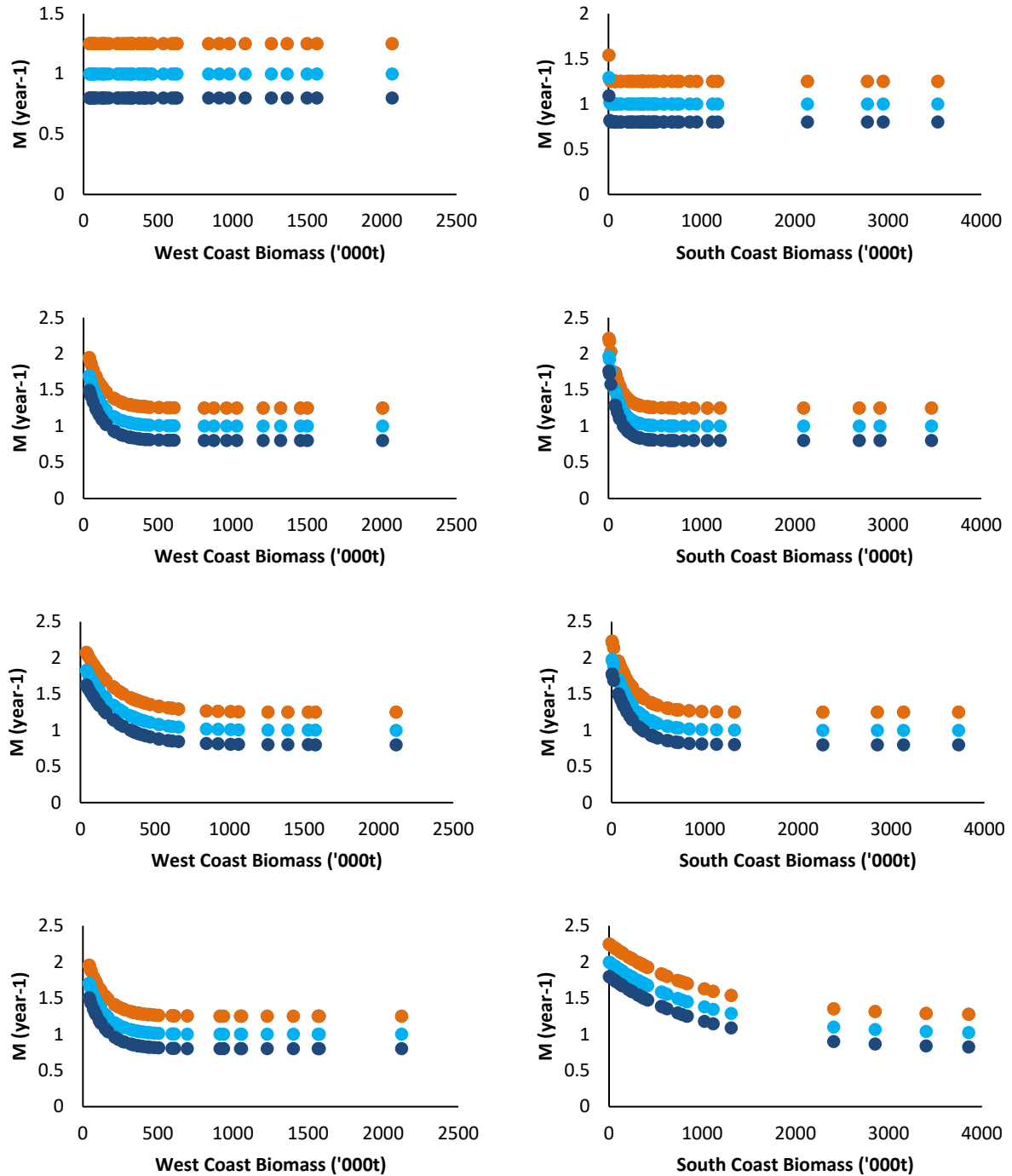


Figure 3. The estimated density dependent relationship between natural mortality and model estimated biomass by coast for $\chi = 0.30$ (top row), $\chi = 0.01$ (2nd row), $\chi = 0.0047$ (3rd row), $\chi_w = 0.009$ and $\chi_s = 0.001$ (4th row). CTS age 0 is shown in light blue, CTS age 1+ is shown in dark blue, WTS age 0 is shown in light orange, WTS age 1+ is shown in dark orange.

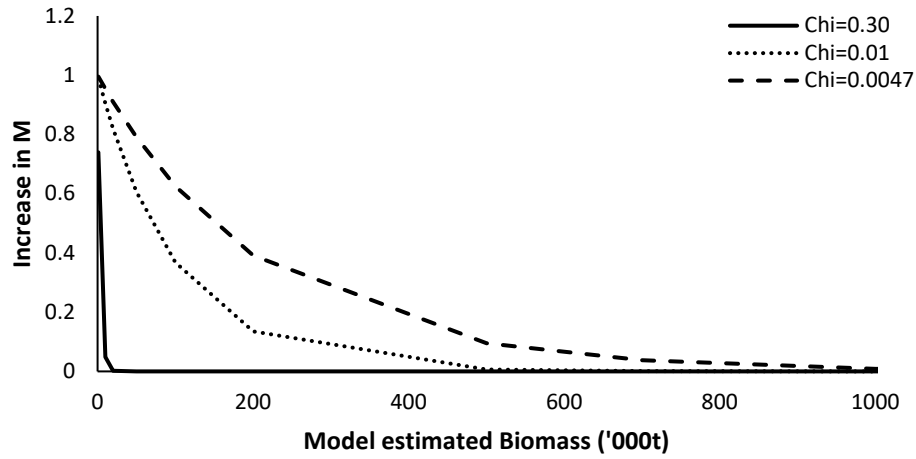


Figure 4. The increase in natural mortality rate from the fixed time invariant rate.

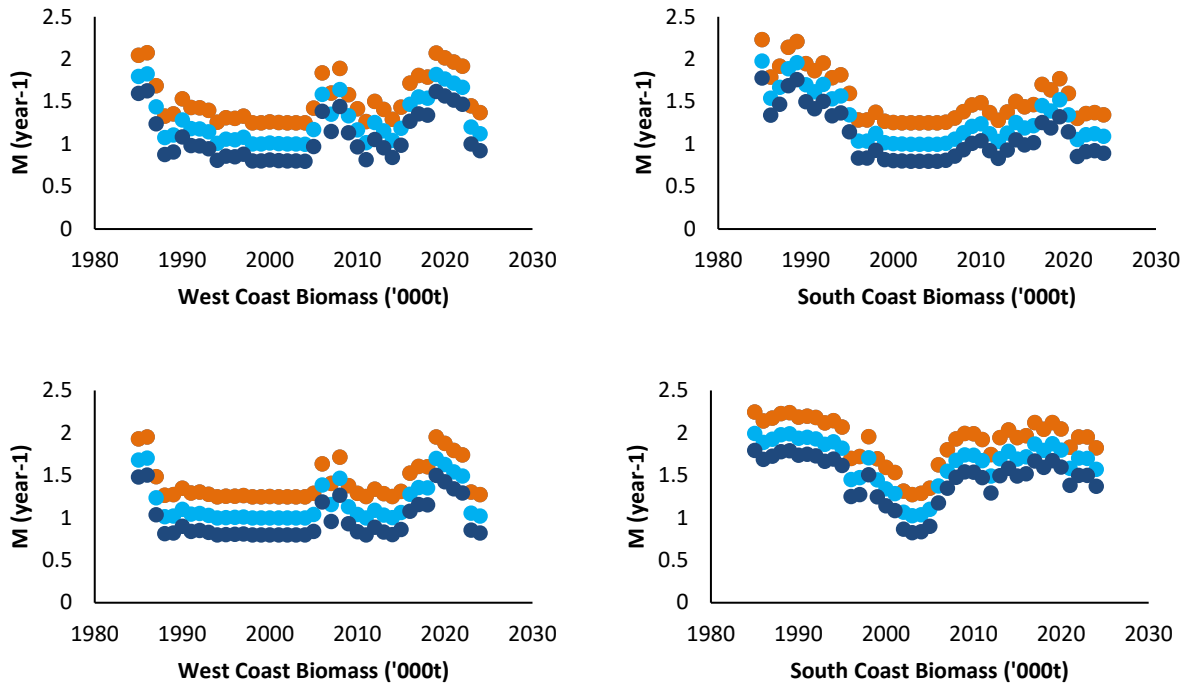


Figure 5. The estimated density dependent natural mortality rate for CTS age 0 (light blue), CTS age 1+ (dark blue), WTS age 0 (light orange) and WTS age 1+ (dark orange) for $\chi = 0.0047$ (top row) and $\chi_w = 0.009$ and $\chi_s = 0.001$ (lower row) against year.

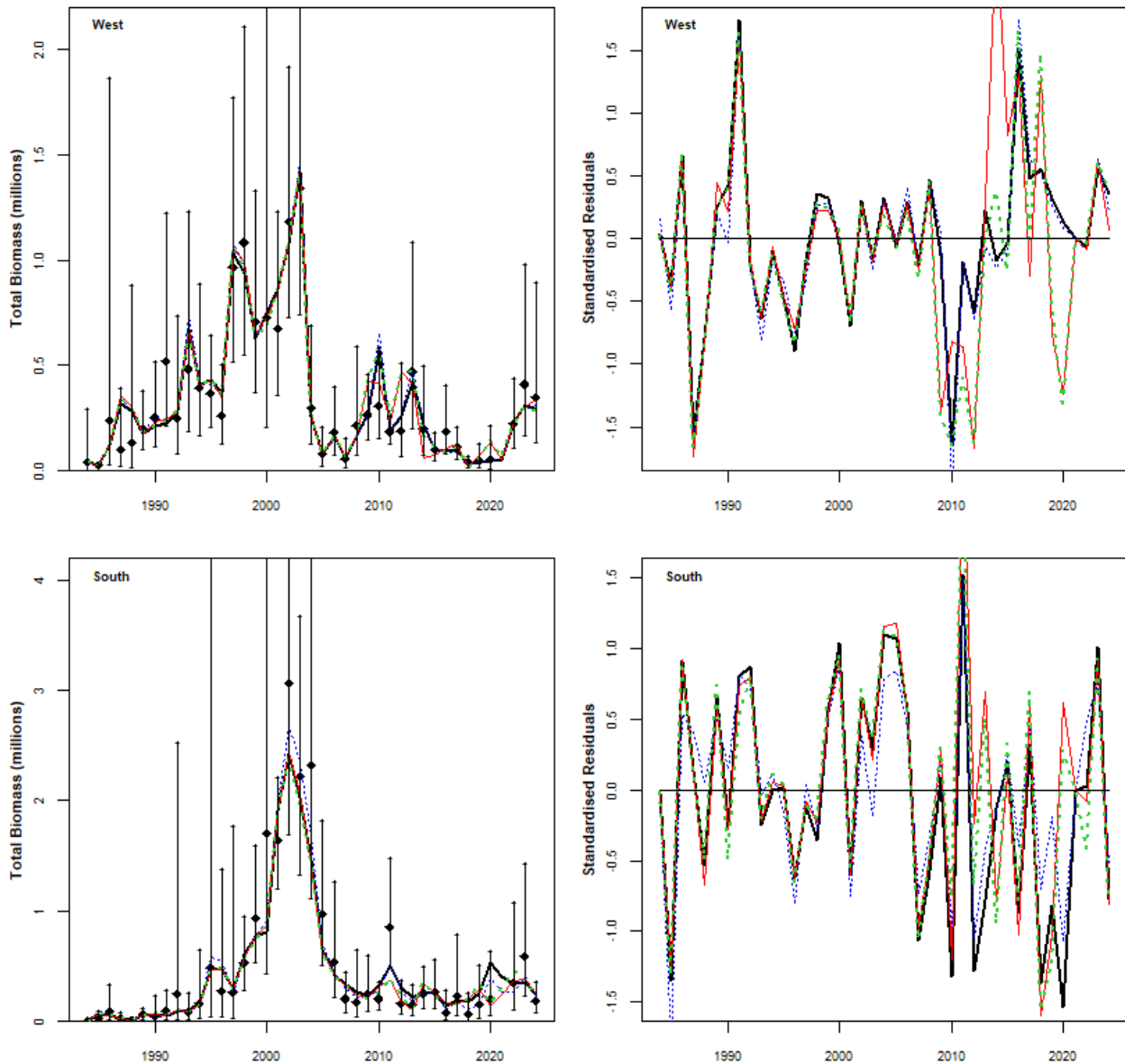


Figure 6. Acoustic survey estimated and model predicted November sardine total biomass from 1984 to 2024 for S_0 (black solid line), S_{DD4} (blue dotted line), S_{prev} (red line) and S_{prev2} (green line). The observed indices are shown with 95% confidence intervals. The standardised residuals (i.e. the residual divided by the corresponding standard deviation, including additional variance where appropriate) from the fits are given in the right hand plots.

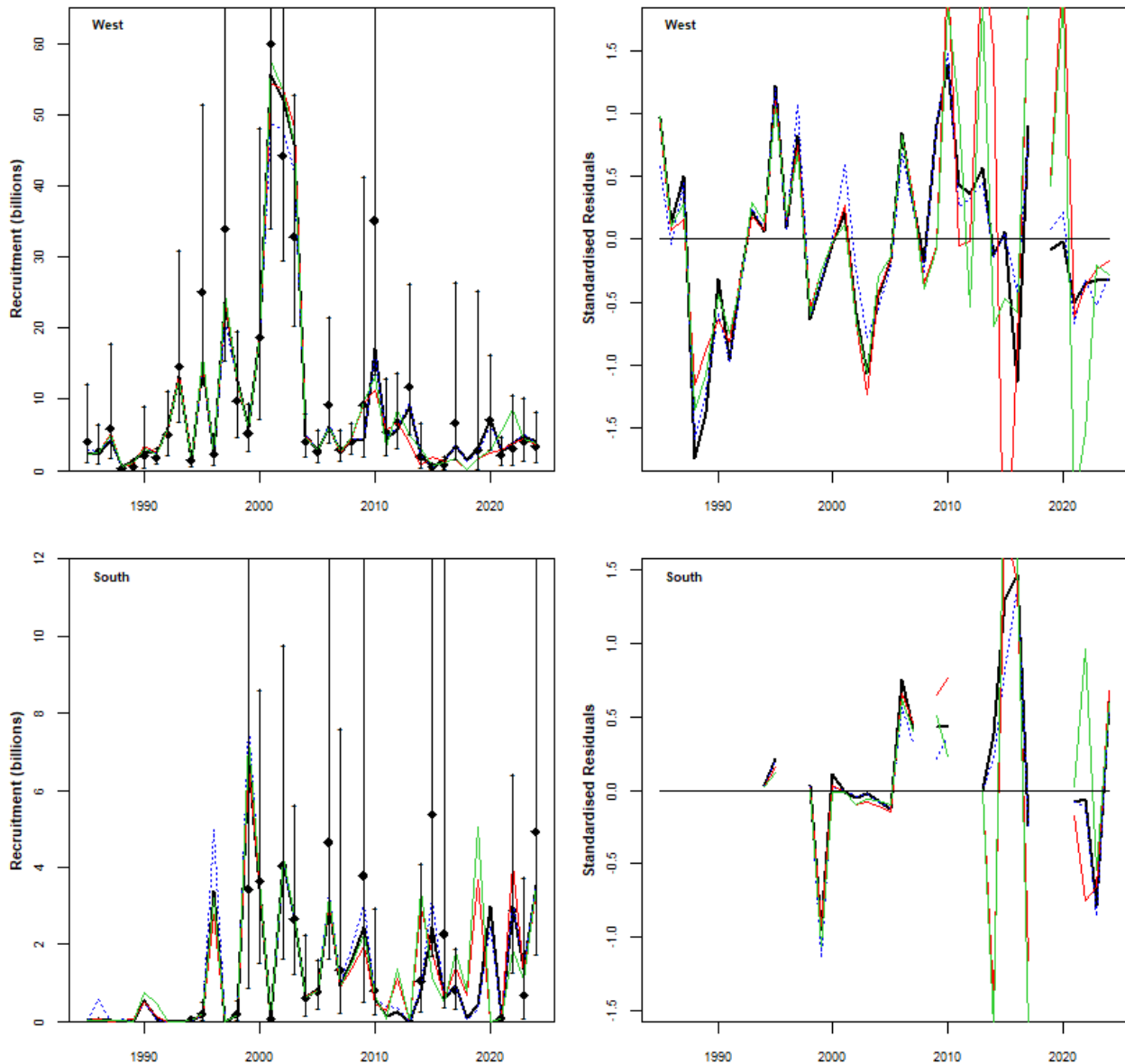


Figure 7. Acoustic survey estimated and model predicted sardine recruitment numbers from May/June 1985 to 2024 for S_0 (black solid line), S_{DD4} (blue dotted line), S_{prev} (red line) and S_{prev2} (green line). There was no survey observation in 2018; the model predicted value corresponds to the recruitment predicted at 8th June 2018 which is the average start date of the survey from 2016, 2017 and 2019 surveys. The survey indices are shown with 95% confidence intervals. The standardised residuals from the fit are given in the right hand plots.

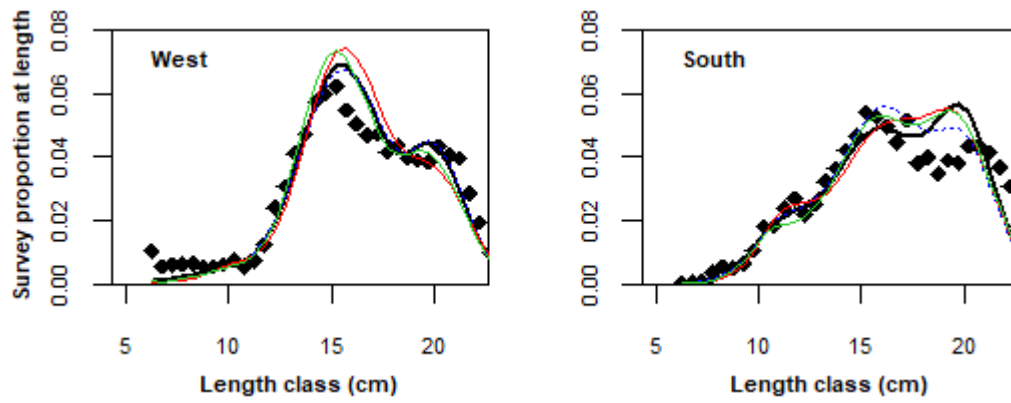


Figure 8. Average (over all years) model predicted and observed proportion-at-length in the November survey for S_0 (black solid line), S_{DD4} (blue dotted line), S_{prev} (red line) and S_{prev2} (green line).

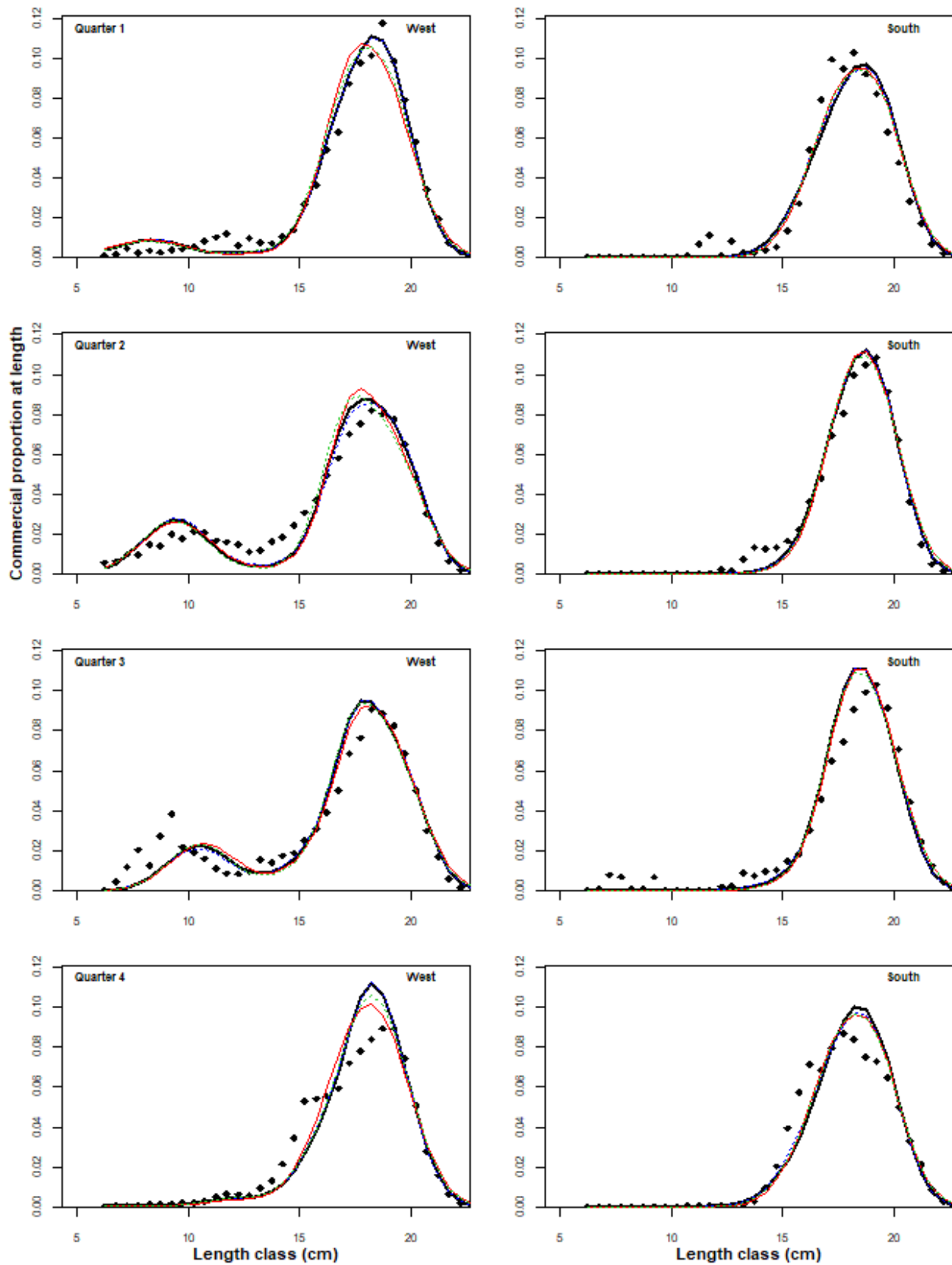


Figure 9. Average (over all years) quarterly model predicted and observed proportions-at-length in the commercial catch for S_0 (black solid line), S_{DD4} (blue dotted line), S_{prev} (red line) and S_{prev2} (green line).

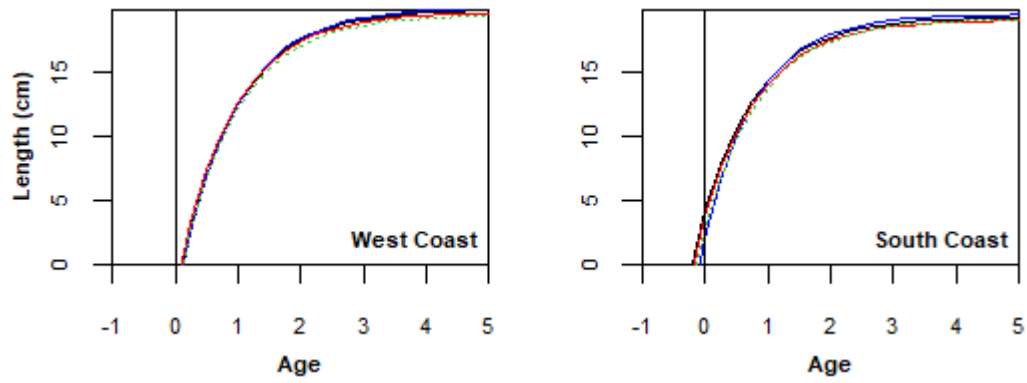


Figure 10. The von Bertalanffy growth curves for S_0 (black solid line), S_{DD4} (blue dotted line), S_{prev} (red line) and S_{prev2} (green line).

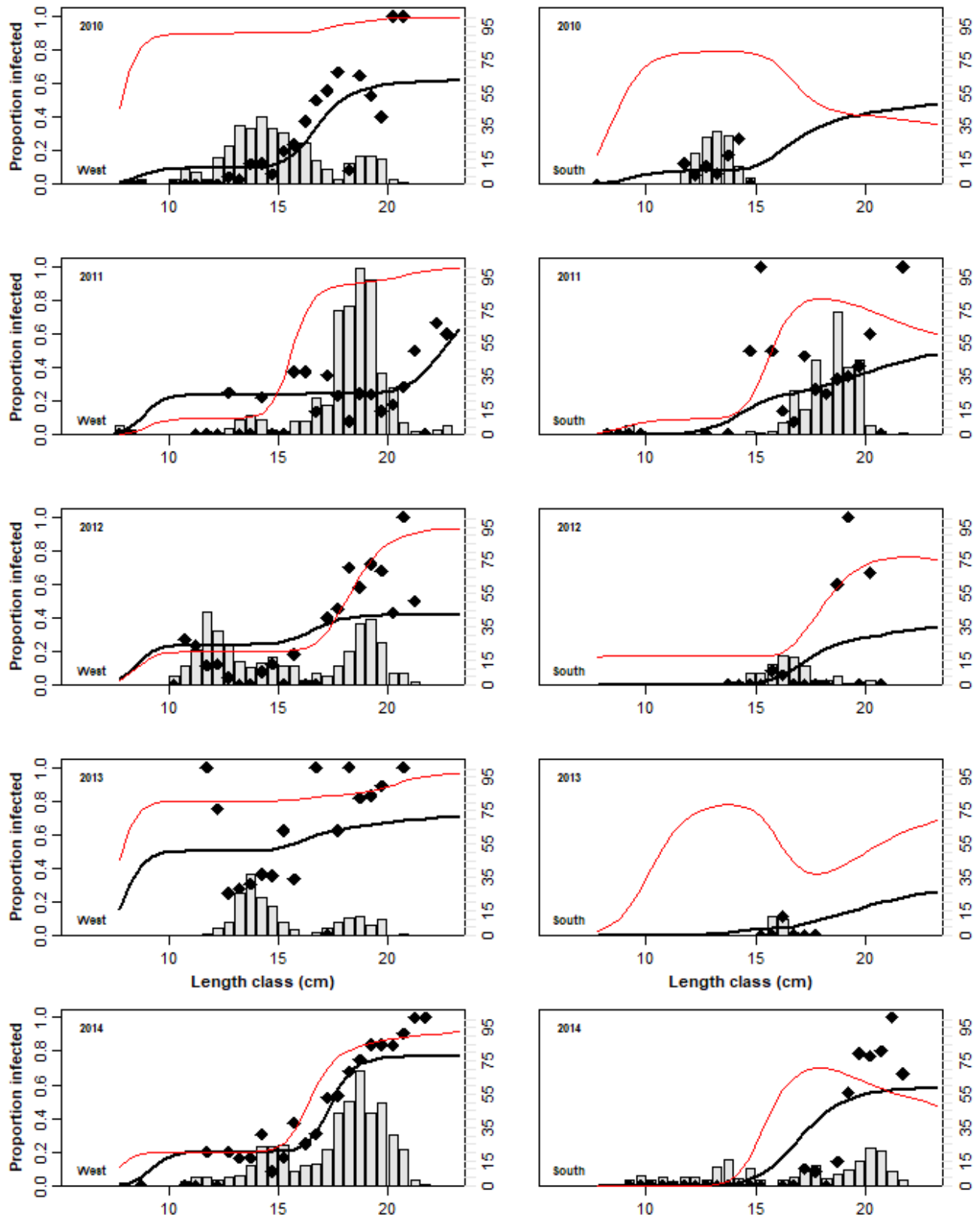


Figure 11. The model estimated proportions-at-length of sardine infected with the parasite (i.e. parasite prevalence-at-length) off the West and South Coasts between 2010 and 2022 together with observed proportions-at-length for S_{prev} . The sample size for each length class is given by the grey bars, plotted against the right vertical axis. The model estimated proportions-at-length for S_0 which does not include these data in the likelihood are shown for interest in red.

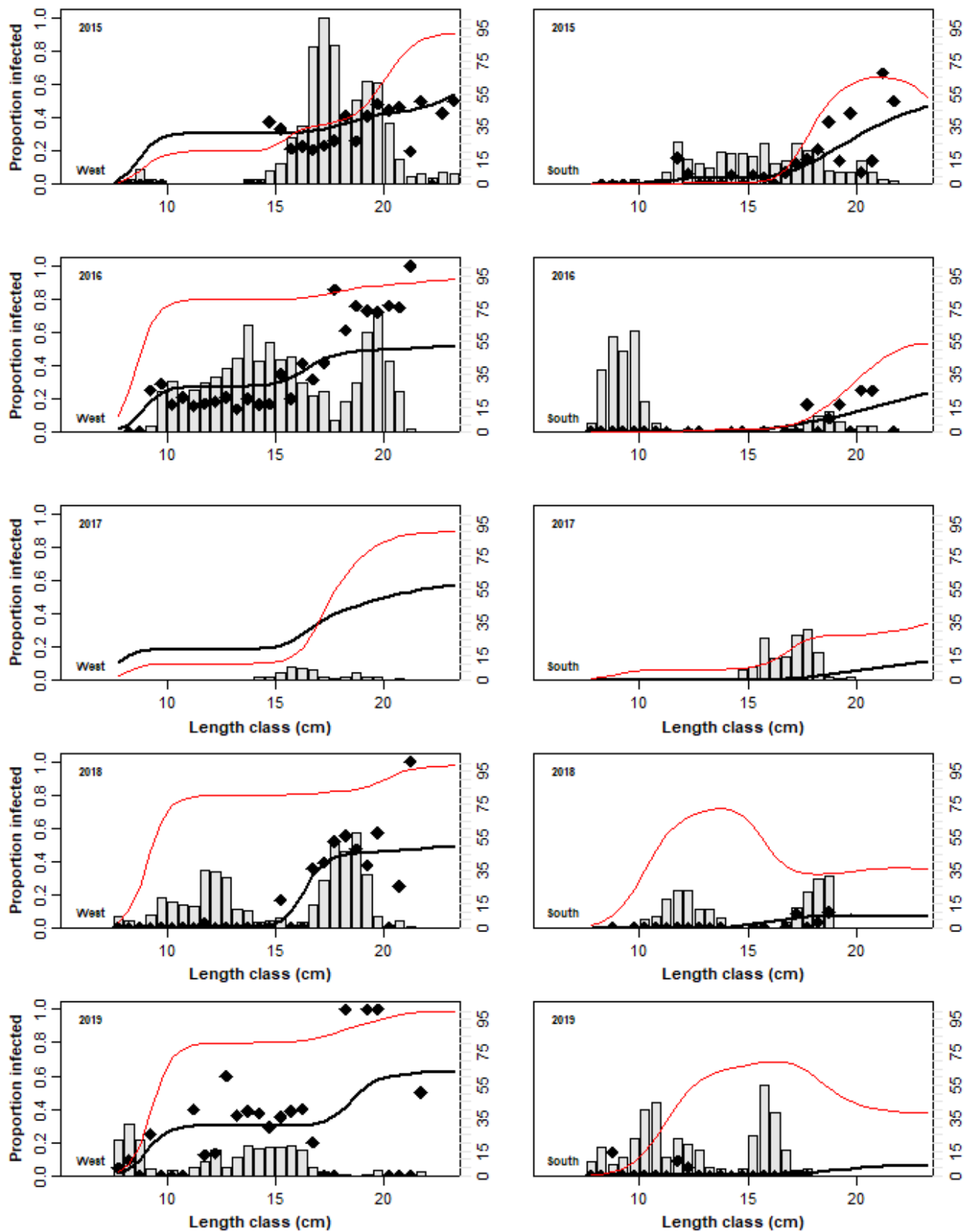


Figure 11 (continued).

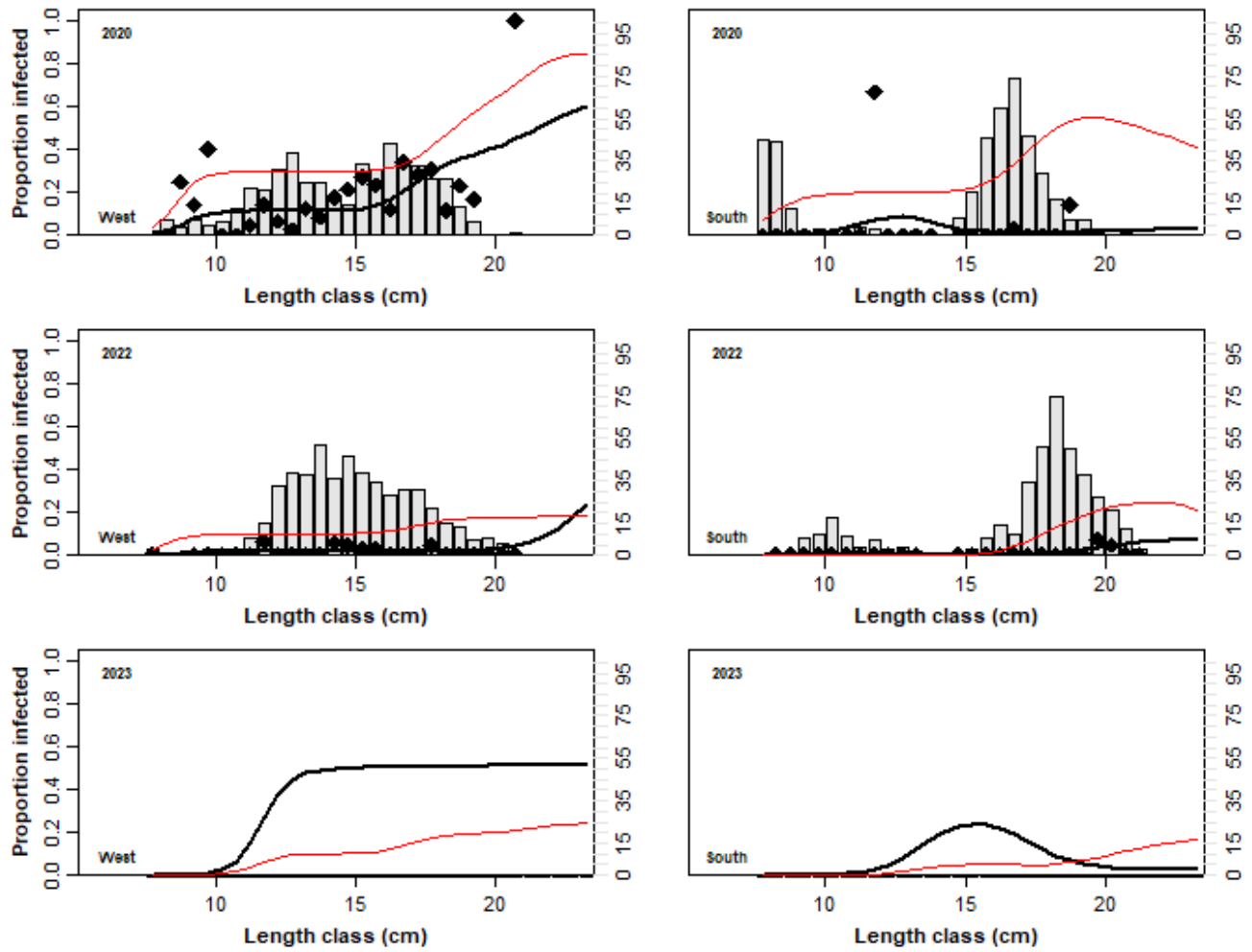


Figure 11 (continued).

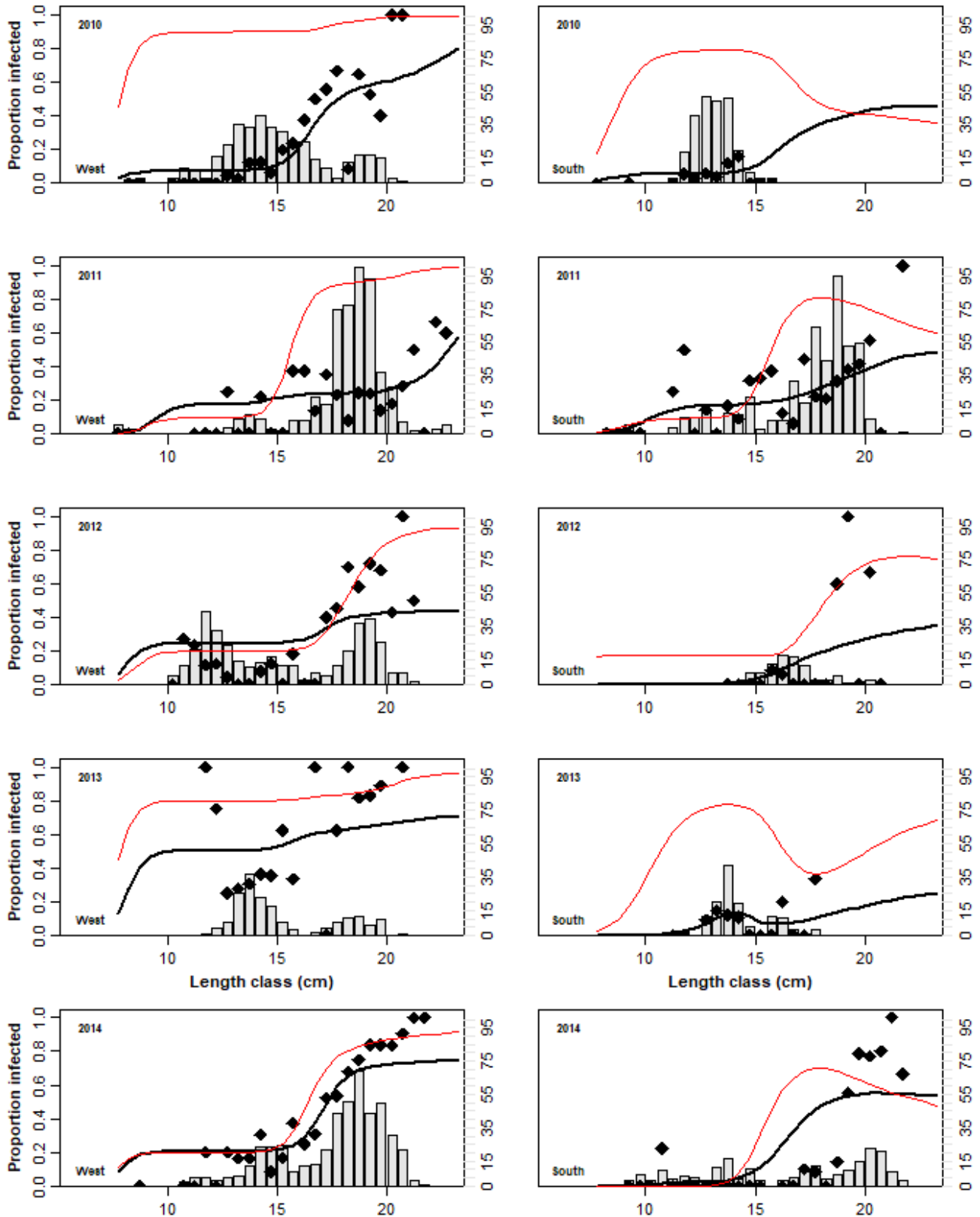


Figure 12. The model estimated proportions-at-length of sardine infected with the parasite (i.e. parasite prevalence-at-length) off the West and South Coasts between 2010 and 2022 together with observed proportions-at-length for S_{prev2} . The sample size for each length class is given by the grey bars, plotted against the right vertical axis. The model estimated proportions-at-length for S_0 which does not include these data in the likelihood are shown for interest in red.

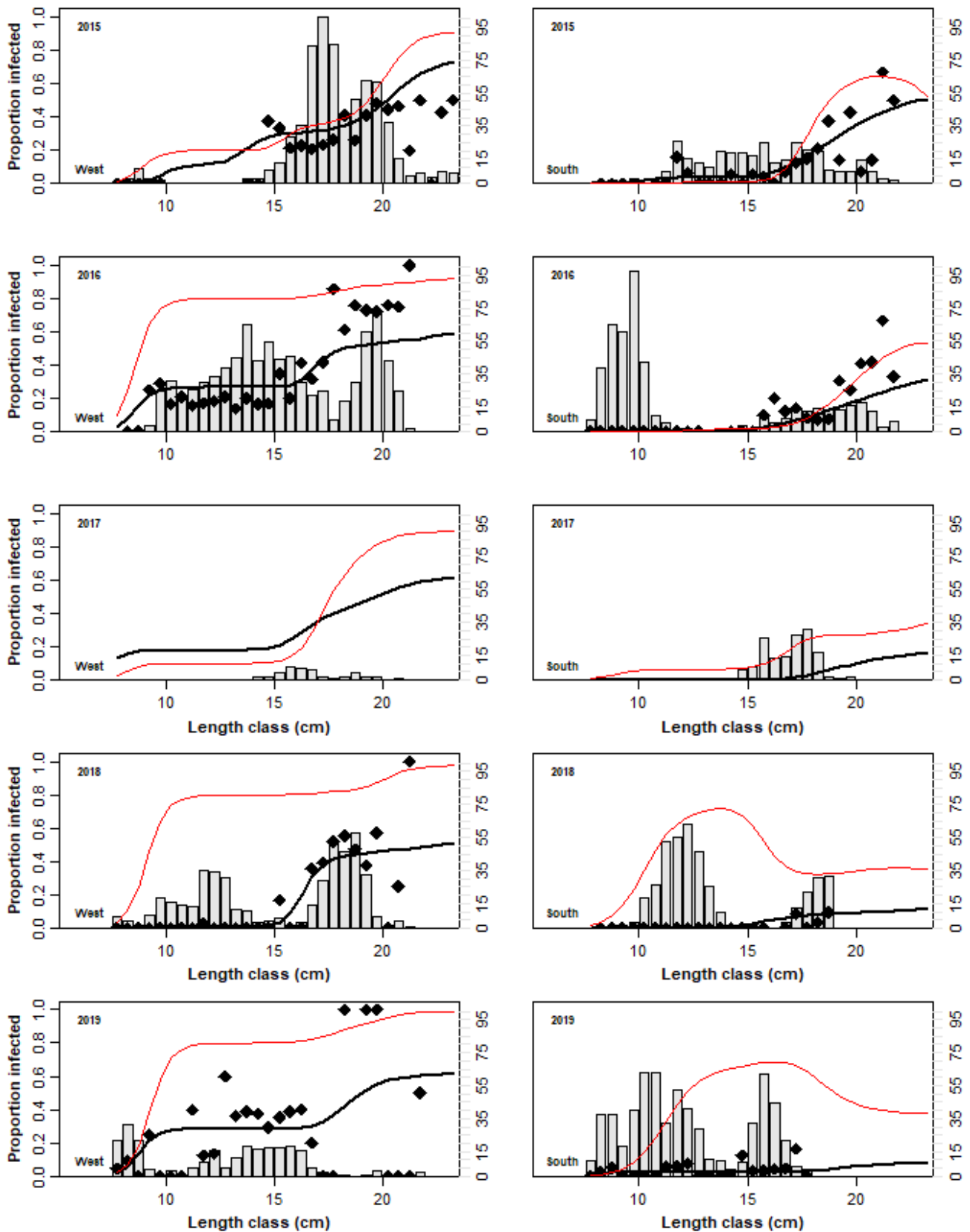


Figure 12 (continued).

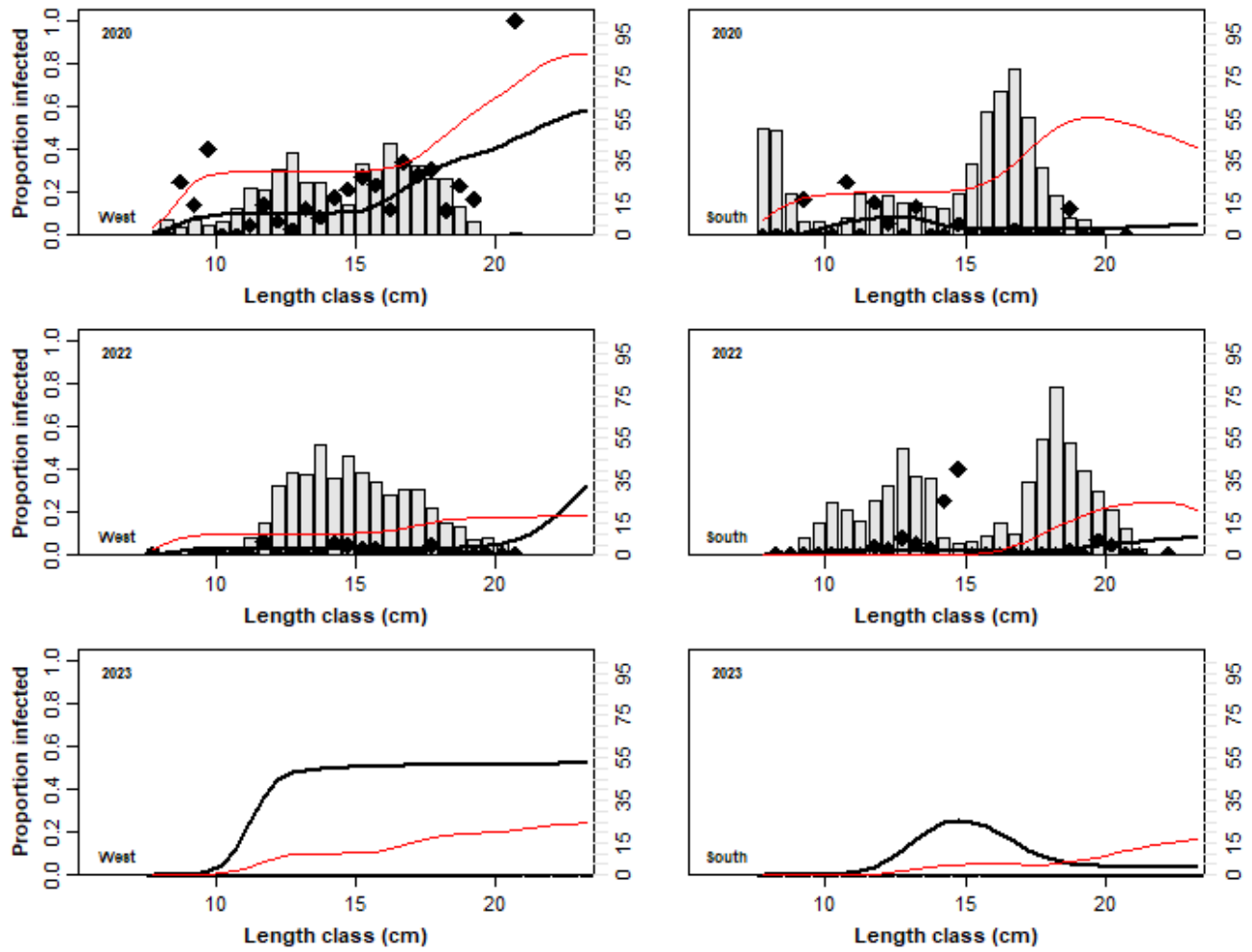


Figure 12 (continued).